Newton's First Law of Motion

This lecture note summarizes Newton's First Law and its applications in daily life.

1. Statement of the Law: An object at rest stays at rest, and an object in motion stays in motion with the same speed and in the same direction, unless acted upon by an unbalanced external force. This is also called the Law of Inertia.

2. Galileo's Thought Experiment: Galileo rolled a ball down one incline and observed that it climbed the opposite incline to nearly the same height. He reasoned that if the second incline were made horizontal, the ball would keep rolling forever, assuming no friction.

This experiment established a new scientific method: deriving physical laws from experimental facts combined with logical reasoning.

3. The Concept of Inertia: Inertia is the tendency of an object to resist changes in its state of motion. All objects have inertia. Mass is the only measure of inertia. The larger the mass, the harder it is to change the motion of the object.

4. Everyday Examples: When a car brakes suddenly, passengers lurch forward because their bodies tend to keep moving. A hammer head can be tightened by striking the handle against a bench, because the heavy head keeps moving when the handle stops.

5. Common Misconception: Force is NOT needed to maintain motion. Force is needed to CHANGE motion. Aristotle's view was corrected by Galileo and Newton after nearly two thousand years.

6. Key Takeaways: The First Law defines inertial frames and reveals the correct relation between force and motion. It is the qualitative foundation on which the Second Law F = ma is built.

Review Questions: State the First Law in your own words. Explain why passengers lean forward when a bus stops suddenly. Why is mass, not volume or weight, the measure of inertia?